

# Comprehensive Mathematical Guide & Formula Compendium

## Analytical Geometry: Straight Lines

Complete Formulas, Conditions, Operational Methods, and Critical Traps

### 1 Foundations of 2D Coordinate Geometry

#### 1.1 Essential Prerequisite Formulas

- **Distance Formula:** The Euclidean distance  $d$  between  $P(x_1, y_1)$  and  $Q(x_2, y_2)$  is:

$$PQ = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

- **Section Formula (Internal Division):** Point  $R(x, y)$  dividing  $PQ$  internally in ratio  $m : n$ :

$$x = \frac{mx_2 + nx_1}{m + n}, \quad y = \frac{my_2 + ny_1}{m + n}$$

- **Section Formula (External Division):** Point  $R(x, y)$  dividing  $PQ$  externally in ratio  $m : n$ :

$$x = \frac{mx_2 - nx_1}{m - n}, \quad y = \frac{my_2 - ny_1}{m - n} \quad (m \neq n)$$

- **Midpoint Formula:** Coordinates of the midpoint  $M$  of segment  $PQ$ :

$$M \left( \frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

- **Centroid of a Triangle:** For a triangle with vertices  $A(x_1, y_1)$ ,  $B(x_2, y_2)$ , and  $C(x_3, y_3)$ :

$$G \left( \frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3} \right)$$

- **Area of a Triangle:** The area  $\Delta$  bounded by  $A(x_1, y_1)$ ,  $B(x_2, y_2)$ ,  $C(x_3, y_3)$  is:

$$\Delta = \frac{1}{2} \left| x_1(y_2 - y_3) + x_2(y_3 - y_1) + x_3(y_1 - y_2) \right| = \frac{1}{2} \begin{vmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{vmatrix}$$

- **Condition for Collinearity:** Three points  $A, B, C$  are collinear if and only if  $\Delta = 0$ .

#### Tricks & Traps: Prerequisites

1. **External Division Ratio Sign:** When applying internal section formula with an unknown ratio  $k : 1$ , if  $k$  evaluates to a negative number ( $k < 0$ ), it signifies that the point divides the line segment *externally*.
2. **Area Determinant Absolute Value:** Always apply the outer absolute value bracket *after* evaluating the expansion. A negative determinant value indicates clockwise vertex ordering, but physical area must be non-negative.

## 2 Slope, Inclination, and Angle Between Lines

### 2.1 Inclination and Slope Definitions

- **Inclination ( $\theta$ ):** The anti-clockwise angle made by a line with the positive direction of the  $x$ -axis, where  $0^\circ \leq \theta < 180^\circ$  ( $0 \leq \theta < \pi$ ).
- **Slope / Gradient ( $m$ ):**  $m = \tan \theta$  ( $\theta \neq 90^\circ$ ).
- **Two-Point Slope Formula:** Slope of a line passing through  $P(x_1, y_1)$  and  $Q(x_2, y_2)$  ( $x_1 \neq x_2$ ):

$$m = \frac{y_2 - y_1}{x_2 - x_1} = \frac{y_1 - y_2}{x_1 - x_2}$$

- **Horizontal Line:**  $\theta = 0^\circ \implies m = \tan 0^\circ = 0$ .
- **Vertical Line:**  $\theta = 90^\circ \implies m = \tan 90^\circ$  is **undefined** ( $\infty$ ).

### 2.2 Parallelism, Perpendicularity, and Intersecting Angles

- **Parallel Lines Condition** ( $l_1 \parallel l_2$ ): Equal inclinations  $\theta_1 = \theta_2 \iff m_1 = m_2$ .
- **Perpendicular Lines Condition** ( $l_1 \perp l_2$ ): Inclinations differ by  $90^\circ \iff m_1 m_2 = -1 \implies m_2 = -\frac{1}{m_1}$ .
- **Acute Angle ( $\theta$ ) Between Two Lines:** For non-vertical lines with slopes  $m_1$  and  $m_2$ :

$$\tan \theta = \left| \frac{m_2 - m_1}{1 + m_1 m_2} \right| \quad (1 + m_1 m_2 \neq 0)$$

The obtuse angle  $\phi$  between the lines is given by  $\phi = 180^\circ - \theta$ .

#### When and How to Use Angle Formulas

Use  $\tan \theta = \left| \frac{m_2 - m_1}{1 + m_1 m_2} \right|$  whenever you are given the angle between two lines (e.g.,  $45^\circ$  or  $60^\circ$ ) and need to find the slope of an unknown line. Removing the absolute value produces **two slopes**, representing the two lines inclined at angle  $\theta$  to the given line.

#### Tricks & Traps: Slopes and Angles

1. **The Vertical Line Trap:** If one of the lines is vertical, its slope  $m = \infty$ . The tangent angle formula fails due to division by zero. Instead, compute inclination directly: if line 1 has inclination  $\theta_1$ , line 2 has inclination  $90^\circ$ , then the angle is  $|90^\circ - \theta_1|$ .
2. **Missing Solution in Square/Triangle Problems:** When finding lines making an angle  $\alpha$  with a line  $l$ , omitting the  $\pm$  sign when opening the absolute value strips away half of the geometric solutions.

## 3 Standard Forms of Equations of a Straight Line

### 3.1 Summary of Standard Forms

### 3.2 Detailed Analysis of Normal Form: $x \cos \omega + y \sin \omega = p$

- $p$ : Length of the perpendicular (normal) drawn from the origin  $(0, 0)$  to the line. Strictly  $p > 0$ .

| Form Name               | Equation                                                        | Given Parameters                                         |
|-------------------------|-----------------------------------------------------------------|----------------------------------------------------------|
| Horizontal Line         | $y = b$ or $y = -b$                                             | $y$ -intercept $b$ or $-b$ ( $m = 0$ )                   |
| Vertical Line           | $x = a$ or $x = -a$                                             | $x$ -intercept $a$ or $-a$ ( $m$ undefined)              |
| Point-Slope Form        | $y - y_1 = m(x - x_1)$                                          | Point $(x_1, y_1)$ and slope $m$                         |
| Two-Point Form          | $y - y_1 = \frac{y_2 - y_1}{x_2 - x_1}(x - x_1)$                | Points $(x_1, y_1)$ and $(x_2, y_2)$                     |
| Slope-Intercept ( $y$ ) | $y = mx + c$                                                    | Slope $m$ and $y$ -intercept $c$                         |
| Slope-Intercept ( $x$ ) | $y = m(x - d)$                                                  | Slope $m$ and $x$ -intercept $d$                         |
| Intercept Form          | $\frac{x}{a} + \frac{y}{b} = 1$                                 | $x$ -intercept $a$ and $y$ -intercept $b$                |
| Normal (Perpendicular)  | $x \cos \omega + y \sin \omega = p$                             | Normal length $p > 0$ , normal angle $\omega$            |
| Parametric / Distance   | $\frac{x - x_1}{\cos \theta} = \frac{y - y_1}{\sin \theta} = r$ | Point $(x_1, y_1)$ , inclination $\theta$ , distance $r$ |

- $\omega$ : Angle made by the perpendicular segment with the positive  $x$ -axis ( $0 \leq \omega < 360^\circ$ ).

### 3.3 Parametric (Distance) Form: Operational Utility

Any point  $P(x, y)$  at a directed distance  $r$  along a line from a fixed point  $(x_1, y_1)$  has coordinates:

$$x = x_1 + r \cos \theta, \quad y = y_1 + r \sin \theta$$

- Use  $+r$  for points in the direction of inclination  $\theta$ .
- Use  $-r$  for points in the opposite direction along the line.

#### Tricks & Traps: Standard Forms

1. **Normal Form Sign Violation:** The distance  $p$  from the origin MUST be strictly positive ( $p > 0$ ). If reduction yields a negative constant on the right side, multiply the entire equation by  $-1$  before determining  $\omega$ .
2. **Intercept Form Division by Zero:** Intercept form  $\frac{x}{a} + \frac{y}{b} = 1$  cannot represent lines passing through the origin ( $a = 0, b = 0$ ) or lines parallel to axes ( $a = \infty$  or  $b = \infty$ ).

## 4 General Linear Equation and Reductions

### 4.1 General Form: $Ax + By + C = 0$

A first-degree polynomial equation in two variables  $x$  and  $y$  always represents a straight line, provided  $A$  and  $B$  are not simultaneously zero ( $A^2 + B^2 \neq 0$ ).

### 4.2 Algorithmic Reductions

1. **Reduction to Slope-Intercept Form ( $y = mx + c$ ):**

$$By = -Ax - C \implies y = \left(-\frac{A}{B}\right)x + \left(-\frac{C}{B}\right) \quad (B \neq 0)$$

$$\text{Slope } m = -\frac{A}{B}, \quad y\text{-intercept } c = -\frac{C}{B}.$$

## 2. Reduction to Intercept Form ( $\frac{x}{a} + \frac{y}{b} = 1$ ):

$$Ax + By = -C \implies \frac{x}{-\frac{C}{A}} + \frac{y}{-\frac{C}{B}} = 1 \quad (A, B, C \neq 0)$$

$$x\text{-intercept } a = -\frac{C}{A}, \quad y\text{-intercept } b = -\frac{C}{B}.$$

## 3. Reduction to Normal Form ( $x \cos \omega + y \sin \omega = p$ ): Divide $Ax + By = -C$ by $\pm\sqrt{A^2 + B^2}$ (choose sign opposite to $C$ so right side becomes positive):

$$\left( \frac{-A}{\pm\sqrt{A^2 + B^2}} \right) x + \left( \frac{-B}{\pm\sqrt{A^2 + B^2}} \right) y = \frac{|C|}{\sqrt{A^2 + B^2}}$$
$$\cos \omega = \frac{-A}{\pm\sqrt{A^2 + B^2}}, \quad \sin \omega = \frac{-B}{\pm\sqrt{A^2 + B^2}}, \quad p = \frac{|C|}{\sqrt{A^2 + B^2}}$$

# 5 Distance Theorems and Formulas

## 5.1 Perpendicular Distance from a Point to a Line

The perpendicular distance  $d$  from point  $P(x_1, y_1)$  to the line  $Ax + By + C = 0$  is:

$$d = \frac{|Ax_1 + By_1 + C|}{\sqrt{A^2 + B^2}}$$

**Special Case:** Distance of origin  $(0, 0)$  from  $Ax + By + C = 0$  is  $d = \frac{|C|}{\sqrt{A^2 + B^2}}$ .

## 5.2 Distance Between Two Parallel Lines

For two parallel lines  $L_1 : Ax + By + C_1 = 0$  and  $L_2 : Ax + By + C_2 = 0$ :

$$d = \frac{|C_1 - C_2|}{\sqrt{A^2 + B^2}}$$

If parallel lines are given in slope-intercept form  $y = mx + c_1$  and  $y = mx + c_2$ :

$$d = \frac{|c_1 - c_2|}{\sqrt{1 + m^2}}$$

### Tricks & Traps: Parallel Line Distances

**The Coefficient Mismatch Trap:** Before applying  $d = \frac{|C_1 - C_2|}{\sqrt{A^2 + B^2}}$ , you MUST ensure the coefficients  $A$  and  $B$  are **identical** in both equations. For example, if given  $2x + 3y + 4 = 0$  and  $4x + 6y + 15 = 0$ , scale the first equation to  $4x + 6y + 8 = 0$  before computing  $|15 - 8|/\sqrt{4^2 + 6^2}$ .

# 6 Advanced Analytical Rules, Foot, Image, and Concurrency

## 6.1 Foot of Perpendicular and Image Formulas

Let  $P(x_1, y_1)$  be a point and  $Ax + By + C = 0$  be a line.

- **Foot of Perpendicular**  $Q(h, k)$  from  $P(x_1, y_1)$  onto  $Ax + By + C = 0$ :

$$\frac{h - x_1}{A} = \frac{k - y_1}{B} = -\frac{Ax_1 + By_1 + C}{A^2 + B^2}$$

- **Image / Reflection**  $R(h, k)$  of point  $P(x_1, y_1)$  in line mirror  $Ax + By + C = 0$ :

$$\frac{h - x_1}{A} = \frac{k - y_1}{B} = -2 \frac{Ax_1 + By_1 + C}{A^2 + B^2}$$

## 6.2 Family of Lines and Concurrency

- **Family of Lines:** The equation of any line passing through the intersection of  $L_1 \equiv A_1x + B_1y + C_1 = 0$  and  $L_2 \equiv A_2x + B_2y + C_2 = 0$  is:

$$L_1 + \lambda L_2 = 0 \implies (A_1 + \lambda A_2)x + (B_1 + \lambda B_2)y + (C_1 + \lambda C_2) = 0 \quad (\lambda \in \mathbb{R})$$

- **Concurrency Condition:** Three lines  $A_ix + B_iy + C_i = 0$  ( $i = 1, 2, 3$ ) are concurrent if and only if:

$$\begin{vmatrix} A_1 & B_1 & C_1 \\ A_2 & B_2 & C_2 \\ A_3 & B_3 & C_3 \end{vmatrix} = 0$$

## 6.3 Position of Points Relative to a Line

Points  $P(x_1, y_1)$  and  $Q(x_2, y_2)$  lie on:

- **Same side** of  $Ax + By + C = 0 \iff (Ax_1 + By_1 + C)(Ax_2 + By_2 + C) > 0$ .
- **Opposite sides** of  $Ax + By + C = 0 \iff (Ax_1 + By_1 + C)(Ax_2 + By_2 + C) < 0$ .

### Equations of Angle Bisectors

The bisectors of angles between lines  $A_1x + B_1y + C_1 = 0$  and  $A_2x + B_2y + C_2 = 0$  are:

$$\frac{A_1x + B_1y + C_1}{\sqrt{A_1^2 + B_1^2}} = \pm \frac{A_2x + B_2y + C_2}{\sqrt{A_2^2 + B_2^2}}$$